

# Yanze Wang - Curriculum Vitae

Academy of Mathematics and Systems Science, University of Chinese Academy of Sciences Ph.D.

Candidate in Pure Mathematics — expected June 2027

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## Research & Practice

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- **Mathematics:** birational geometry and the minimal model program — Sarkisov program, algebraically integrable foliations, moduli spaces.
- **AI practice:** local small projects with PyTorch / libtorch; agent-framework building.
- **AI4Math:** human verification of AI-generated mathematical results; research harnesses and Lean 4 formalization.
- **Engineering:** torch (Python & C++); Rust, C, LaTeX; familiar with Linux and Git; personal CLI / TUI programs developed with vibe coding (Rust).

## Publications

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### Sarkisov Program for Algebraically Integrable Adjoint Foliated Structures.

2026

Yifei Chen, Jihao Liu, and Yanze Wang. *International Mathematics Research Notices*, 2026(6): *rnag045*.

By applying the theory of the minimal model program for adjoint foliated structures, we establish the Sarkisov program for algebraically integrable foliations on klt varieties: any two Mori fiber spaces of such a structure are connected by a sequence of Sarkisov links. Combining with a result of R. Mascharak, we establish the Sarkisov program for foliations in dimension at most 3 with mild singularities. Log and adjoint-foliated versions of the aforementioned Sarkisov programs are also established.

### Flop between algebraically integrable foliations on potentially KLT varieties.

2025

Yifei Chen, Jihao Liu, and Yanze Wang. *International Journal of Mathematics*, 36(11):2550035.

We prove that for any two minimal models of an lc algebraically integrable foliated triple on potentially klt varieties, there exist small birational models that are connected by a sequence of flops. In particular, any two minimal models of lc algebraically integrable foliated triples on  $\mathbb{Q}$ -factorial klt varieties are connected by a sequence of flops. We also discuss the connection between minimal models for possibly non-algebraically integrable foliations on threefolds, assuming the minimal model program for generalized foliated quadruples.

### A Note on the Sarkisov Program.

2025

Yifei Chen and Yanze Wang. *Higher Dimensional Algebraic Geometry: A Volume in Honor of V. V. Shokurov*, LMS Lecture Note Series, pp. 231–263, Cambridge University Press.

This note introduces three methods of the Sarkisov program, which factorizes birational maps of log Mori fibre spaces into elementary Sarkisov links.

## Preprints

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### A klt generalized pair with infinitely generated canonical ring (AI-generated, human-verified).

2026

Jihao Liu and Yanze Wang. *arXiv:2608.03258*.

We construct a projective klt generalized pair  $(X, B, M)$  with infinitely generated generalized log canonical ring  $R(X, K_X + B + M)$ . This gives the first example of a klt variety  $X$  with  $-K_X$  nef whose anticanonical ring  $R(X, -K_X)$  is not finitely generated.

**Twelve common flex lines in a general pencil of cubics** (AI-generated, human-verified). 2026  
Jihao Liu and Yanze Wang. *arXiv:2607.26396*.

We prove that a general pencil of plane cubics over  $\mathbb{C}$  has exactly 12 common flex lines. This answers a question of Ciliberto, Miranda, and Roe.

**A counterexample to the odd-dimensional rank bound for abelian p-group actions** (AI-generated, human-verified).

Jihao Liu and Yanze Wang. *arXiv:2607.04891*.

We give a counterexample to the odd-dimensional extension of the rank bound for faithful abelian p-group actions on smooth Calabi-Yau varieties, conjectured by Moraga.

## Theses

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**M.S.:** “Sarkisov program for foliated pairs”

The minimal model program seeks a good representative in each birational equivalence class; these representatives split into minimal models and Mori fiber spaces. The thesis introduces the three methods of the Sarkisov program and establishes it for foliated pairs — by reducing F-dlt foliated pairs to klt pairs, one obtains a weak decomposition of birational maps between foliated Mori fiber spaces.

**B.S.:** “Moduli space of curves”

This report constructs moduli spaces of curves and their compactifications via geometric invariant theory (GIT) and stack theory — introducing smooth and DM stable curves, developing the necessary tools (category theory, stack theory, Hilbert scheme, GIT), constructing two kinds of curve moduli spaces, and closing with further questions.

## Invited Talks

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Speaker, **Sarkisov Program for Algebraically Integrable and Threefold Foliations**, Xi'an 2026  
Jiaotong-Liverpool University.

## Projects

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### Programming

**suckless** — personal fork of dwm / dwmblocks (X11 dynamic tiling window manager, C), patched and maintained by hand, no AI assistance.

**Calman** — terminal task & event manager (CLI + TUI, Rust). JSONL/ICS storage, CalDAV-compatible; syncs across devices via a self-hosted Radicale server, delegating to git / vdirsynchroner / rclone.

**Markerss** — TUI RSS reader (Rust, ratatui + feed-rs): Markdown export, URL-only bookmarks.

**Lichtung** — Hugo theme (Go templates / SCSS) built with conversational AI-assisted development; powers two personal sites (docs & blog).

**mumble** — message board (JavaScript, Cloudflare Worker): deploy standalone or embed.

### AI Practice

**Model training:** local practice — pytorch-cpp, llm-from-scratch, build-llm-from-zero, tiny-reasoning-rl-lab, etc.

**AiKit** — agent framework: agents / skills / plugins for coding agents (pi, opencode, codex, claude code, ...), including pi extensions (rtk, cave, toon, doc, role) and quick-deploy scripts.

**Open-source:** forks, improvements & usage — [Rethlas](#) (natural-language math reasoning), co-mathematician, etc.; contribution to the Lean 4 platform [prove2.me](#).

**Closed beta:** agentero, math research workstation & paper reader (temporarily closed-source).

## Education

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### Ph.D.

Mathematics PhD (in progress), AMSS, University of Chinese Academy of Sciences — expected June 2027.

*Advisor:* Prof. Yifei Chen.

*Research:* birational geometry — Sarkisov program, algebraically integrable foliations.

### M.S.

Mathematics M.S., AMSS, University of Chinese Academy of Sciences.

*Advisor:* Prof. Yifei Chen.

*Thesis:* “Sarkisov program for foliated pairs”.

### B.S.

Mathematics B.S., Beihang University.

*Thesis:* “Moduli space of curves”.

Undergraduate teaching assistant: Linear Algebra, Algebraic Geometry.

- MCM/ICM — Meritorious Winner (twice)
- Hua Luogeng Scholarship, Beihang University (twice)